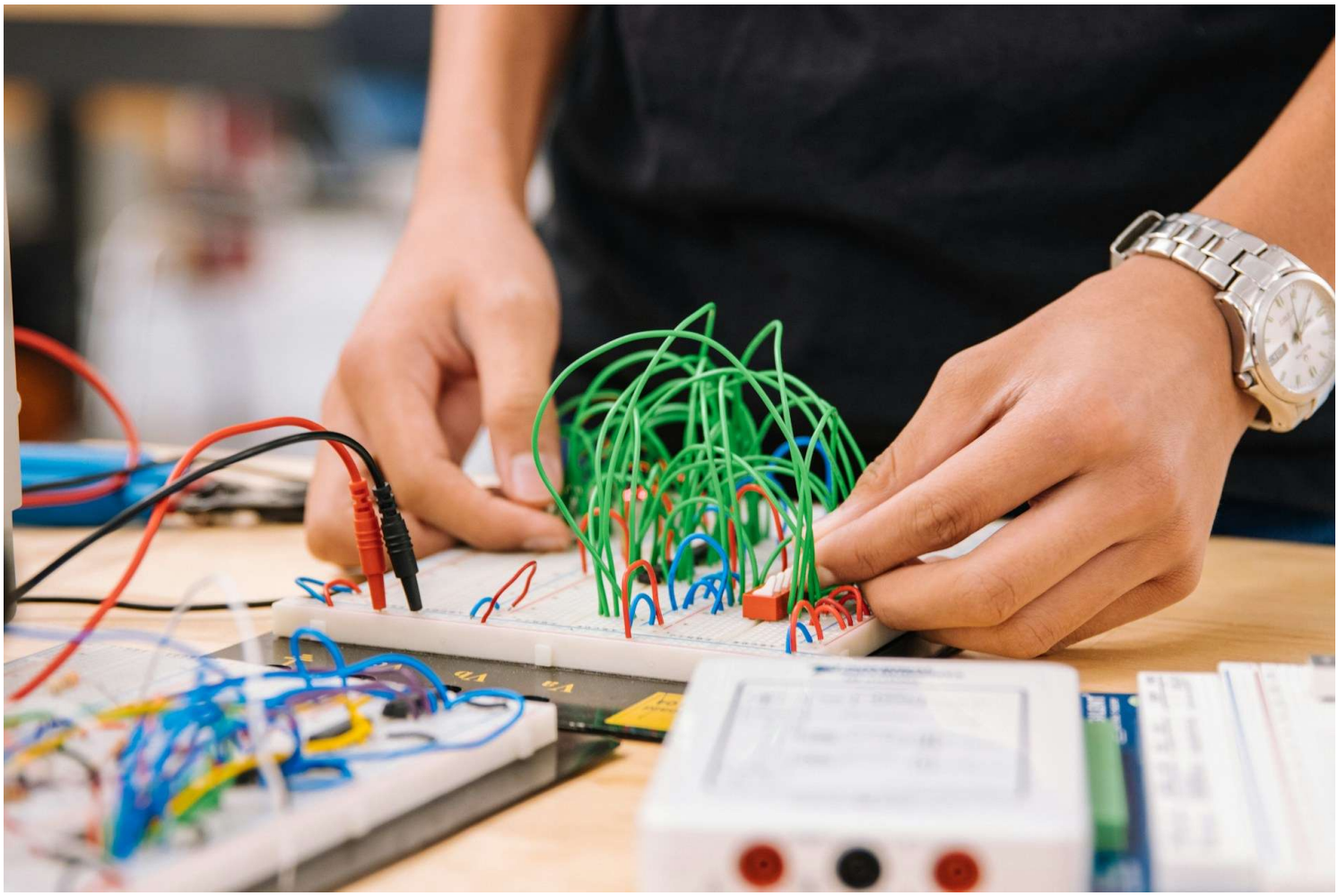
Services

1. Home

Services
Learning from the Best Experiences

At Advanced Studies in Engineering, we strive to provide our clients with the best in-class technical consulting services for advanced scientific studies.



Scientific competence

Qualified advanced scientific lecturers across multiple engineering domains related to electricity.

Industry-Grade Tools

Using advanced engineering software such as PSCAD, PSS/E, and PLS-CADD.

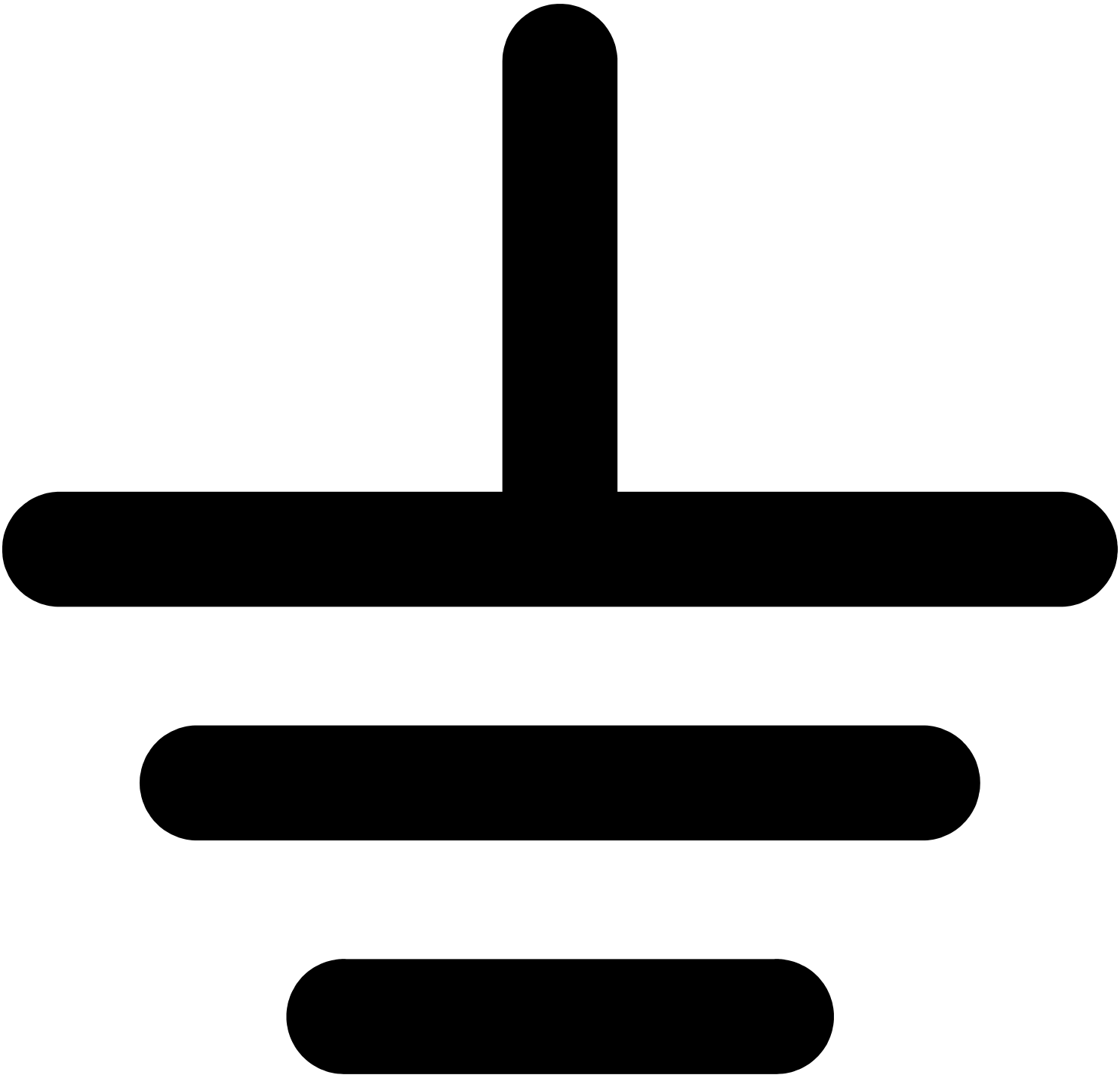
Studies
System Impact Issues/study

Cable Energization

Identify and mitigate frequency disturbances to protect power equipment

TOV – Resonance

Parallel, serial and ferrosonance



Ground faults and clearing

Analyze fault currents to ensure safe system clearing

System Islanding

Build islanding schemes for reliable power network operation.

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Generator self-excitation

Analyze capacitive loading risks to prevent voltage rise.

Transient Stability

Evaluate grid response to sudden disturbances for system recovery

Equipment Design Issues/study

Cable Energization

Evaluating inrush currents and switching transients for safe cable activation

Auto-reclosing Overvoltage

Handling voltage spikes during high-speed automatic grid restoration

Ground faults and clearing

Assessing stored energy release to ensure safe cable system operation.

Zero-miss effect

Analyzing delayed current zero-crossings to ensure safe circuit interruption

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Analyzing delayed current zero-crossings to ensure safe circuit interruption

Generator self-excitation

Mitigating interference between power lines and nearby metallic structures.

System Planning Issues/study

Reactive Power Compensation*

Optimizing power factor to ensure efficient grid voltage stability

Ferranti Effect*

Mitigating voltage rise on long, lightly loaded transmission lines

Active Power Transmission (limits and contingencies)*

Defining safe transfer limits to ensure reliable power grid flow

System technical issues and study tools Issues

Study impact issues using EMT for DC-kHz

- Harmonic frequency scans
- TOV – Ground faults and clearing
- TOV – Resonance
- TOV – System Islanding and load rejection
- Ferrosonance

Issues

Study equipment design issues using EMT

- Cable energization – DC-kHz
- Auto-reclosing overvoltages – DC-kHz
- Discharge of the cable – DC-kHz
- Shealth overvoltages and protection – DC-kHz
- DC offset / zero-miss effect – DC-kHz
- Current transformer saturation (CT) – DC-kHz
- Influence on substation surges and TRV – MHz
- Lightning overvoltages – MHz
- Shunt reactor restrike – MHz

Issues

Learn technical tools to study real planning issues at 50 / 60Hz

- Reactive power compensation – Power flow
- Ferranti effect – Power flow
- Active power transmission (limits and contingencies) – Power flow
- Phase unbalance and transposition – Cable const.

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TESTIMONIALS
Testimonials that reflect our consistent efforts

The feedback from our clients speaks volumes about the quality of our services. Read how we've helped them with reliable solutions.

“

ASE delivered highly detailed power system studies that helped us make informed decisions on our transmission project. Their technical expertise and attention to detail were outstanding.

Paula Sitoe

“

Their team provided reliable and data-driven analysis that significantly improved our system planning process. We highly value their professionalism and technical rigor.

Paulo José

Grid Planning Specialist

Experience with technologies and methodologies used by leading energy companies such as: O

gigawatt Moçambiqu e, sa.

HIDROELÉCTRICA DE CAHORA BASSA

ELECTRICIDADE DE MOÇAMBIQUE, E.P.

sasol

PUMA ENERGY

TotalEnergies

FREQUENTLY ASKED QUESTION

Common questions about our services

Find answers to some of the most frequently asked questions about our power system consulting services, simulation studies, and professional training programs.

What services does ASE provide?

We provide specialized engineering consulting and simulation studies for power systems, including modeling, system analysis, and transmission line design using industry-standard tools such as PSCAD, PSS®E, and PLS-CADD.

Who can benefit from our training programs?

Our training programs are designed for power system engineers, utility professionals, researchers, and students who want to gain practical skills in power system modeling and simulation.

What software tools are used in our training and studies?

We use industry-standard engineering tools including PSCAD for time-domain simulations, PSS®E for power system network analysis, and PLS-CADD for transmission line design and analysis.

How long are the training programs?

Each training level typically runs for **4 weeks**, with sessions held on Saturdays. The program includes lectures, practical simulation exercises, and real-world case studies.

How can I register for a training program?

You can register by contacting us through the website contact form or by sending us an email. Our team will provide details about upcoming training sessions and registration procedures.

Engineering consulting and training services for reliable and efficient power systems. Contact us to learn more.

Company

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PLS-CADD

PSCAD EMTCD

PSS/E

Contact Us

Rua dos Desportistas, nº 921, Maputo, Mozambique.

info@advancedstudies.co.mz

+258 84 123 4567

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